



F13 TECHNOLOGIES

INTERNSHIP PROJECT REPORT

Automatic Payroll Processing & Payslip Delivery Engine



SUBMITTED BY

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UNDER THE GUIDANCE OF

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MENTORS

YEAR

2026-27

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to F13 Technologies for providing us with the opportunity to undertake our internship and work on the “Automatic Payroll Processing and Payslip Delivery Engine.” The project provided valuable practical exposure to cloud computing, serverless architecture, event-driven processing, database design, document generation, secure object storage, and automated email delivery.

We would like to express our heartfelt gratitude to Mr. Kabir Rao and Mr. Nikhil Kumar, our mentors at F13 Technologies, for their continuous guidance, encouragement, technical expertise, and constructive feedback throughout the project. Their mentorship and valuable suggestions played a significant role in the successful completion of this project.

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Chapter 1 – Introduction

1.1 Project Overview

Organizations frequently rely on spreadsheets, manual calculations, and individual email processes to prepare monthly payroll and distribute payslips. These approaches are time-consuming and can introduce calculation errors, duplicate processing, inconsistent records, delayed payslip delivery, and limited auditability.

The Automatic Payroll Processing and Payslip Delivery Engine is a serverless payroll automation solution developed using Amazon Web Services (AWS). The solution stores employee and payroll inputs in Amazon DynamoDB, distributes employee payroll jobs through Amazon SQS, processes each job using AWS Lambda, generates a PDF payslip using ReportLab, stores the payslip in Amazon S3, creates temporary pre-signed access, and delivers the link through Amazon SES.

1.2 Background

Payroll is a critical HR function because salary calculations directly affect employees and organizational financial records. Automating recurring payroll operations can reduce repetitive administrative effort, improve consistency, and create an auditable record of each payroll run.

1.3 Problem Statement

- Manual salary calculation using spreadsheets.
- Risk of calculation errors and inconsistent records.
- Time-consuming monthly payroll preparation.
- Manual payslip generation and distribution.
- Difficulty tracking payroll history.
- Duplicate processing when a queue message is retried.
- Security concerns when payroll documents are shared without controlled access.
- Limited scalability as the employee population grows.

1.4 Existing System

In a traditional process, HR personnel collect salary information, calculate allowances and deductions, prepare salary statements, generate payslips, and distribute them manually. Payroll information may be spread across spreadsheets and folders, making the process difficult to audit and scale.

1.4.1 Limitations

- High manual effort
- Spreadsheet dependency
- Calculation and data-entry errors
- Manual document generation
- Manual email distribution
- Difficult audit trail
- Limited scalability
- Security risks

1.5 Proposed System

The proposed system automates the payroll pipeline using managed AWS services. A payroll request is represented as queue work. The payroll-worker Lambda retrieves employee, salary, and deduction information from DynamoDB, calculates gross salary and deductions, generates the payslip PDF, uploads it to a private S3 bucket, generates temporary access, sends the payslip email using SES, and records the result in DynamoDB.

1.6 Objectives

- Automate monthly payroll processing.
- Calculate salary components and deductions consistently.
- Process employees independently through an SQS-based worker model.
- Generate professional PDF payslips.
- Store payslips securely in private S3.
- Deliver payslip access through SES.
- Maintain payroll history in DynamoDB.
- Provide logging and troubleshooting through CloudWatch.
- Demonstrate scalable and serverless AWS architecture.

1.7 Scope

The project covers employee salary data, deductions, payroll calculation, PDF generation, secure payslip storage, temporary document access, email delivery, queue-based processing, database record keeping, and operational logging. The implementation uses sample employee data and simplified payroll tax rules suitable for an internship/portfolio demonstration. Full statutory payroll compliance, attendance integration, and enterprise HRMS integration are outside the present scope.

1.8 Key Features

- Employee master data
- Salary components
- Deduction configuration
- SQS payroll queue
- Lambda worker
- ReportLab PDF generation
- Private S3 payslip storage
- Pre-signed URL
- SES email delivery
- DynamoDB payroll history
- CloudWatch logging
- IAM-controlled access

Chapter 2 – Literature Survey

2.1 Cloud Computing

Cloud computing provides on-demand access to computing, storage, databases, networking, and application services. It enables rapid deployment and reduces the need to maintain physical infrastructure.

2.2 Serverless Computing

Serverless computing allows application teams to focus on business logic while the cloud platform manages the underlying compute infrastructure. AWS Lambda is well suited to event-driven workloads such as queue-based payroll processing.

2.3 Payroll Automation

Payroll automation applies defined salary and deduction rules to recurring payroll inputs and produces standardized payroll records and payslips. Automation can reduce repetitive work and improve traceability.

2.4 Event-Driven Architecture

An event-driven architecture separates job creation from job execution. In this project, Amazon SQS acts as the queue and AWS Lambda acts as the worker. This design allows employee payroll jobs to be processed independently and supports retry-based failure handling.

2.5 Secure Document Delivery

Payroll payslips contain sensitive information. A private S3 bucket combined with a temporary pre-signed URL provides controlled access to a specific payslip without making the entire bucket public.

2.6 Comparative Analysis

Feature	Traditional Payroll	Proposed AWS System
Calculation	Manual spreadsheet	Lambda-based automation
Processing	Sequential/manual	SQS + Lambda
Database	Files/spreadsheets	DynamoDB
Payslip generation	Manual	ReportLab PDF
Storage	Local/shared folders	Private S3
Delivery	Manual email	SES
Document access	File sharing	Pre-signed URL
Audit	Scattered records	DynamoDB payroll_runs
Infrastructure	Servers/local systems	Serverless AWS

Chapter 3 – System Requirements

3.1 Functional Requirements

- Store employee master data and status.
- Store salary components such as base salary, HRA, transport allowance, and other allowances.
- Store deduction inputs including PF percentage, TDS rate, and loan deduction.
- Accept payroll jobs containing run ID, employee ID, and month.
- Retrieve and validate employee, salary, and deduction records.
- Calculate gross salary, PF, professional tax, TDS, loan deduction, total deductions, and net pay.
- Write payroll results to the payroll_runs table.
- Generate a formatted PDF payslip.
- Upload the PDF to private S3 storage.
- Generate a temporary pre-signed URL.
- Send the payslip URL through Amazon SES.
- Handle failed queue processing using retries/DLQ.
- Record execution details in CloudWatch Logs.

3.2 Non-Functional Requirements

Performance

Employee payroll jobs should be independently processable.

Scalability

SQS and Lambda provide a scalable worker pattern.

Reliability

Queue retries and a dead-letter queue support failure recovery.

Security

IAM controls access, S3 remains private, and temporary URLs limit document exposure.

Auditability

Payroll results and payslip references are retained in DynamoDB.

Maintainability

The Lambda source is organized into calculation, database, payroll, payslip, and tax-rule modules.

Cost Effectiveness

Managed/serverless services reduce the need for continuously running servers.

3.3 Hardware Requirements

Component	Specification
Development system	Modern desktop/laptop
Memory	8 GB recommended
Internet	Stable connection

Browser	Modern web browser
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3.4 Software and AWS Requirements

Component	Purpose
Python 3.14	Lambda runtime used by payroll-worker
Boto3	AWS SDK for Python
ReportLab	PDF generation
AWS Lambda	Payroll worker
Amazon SQS	Queue
Amazon DynamoDB	Payroll database
Amazon S3	Payslip storage
Amazon SES	Email delivery
Amazon CloudWatch	Logs
AWS IAM	Access control
AWS Region	Asia Pacific (Hyderabad), ap-south-2

Chapter 4 – System Architecture

4.1 Architecture Overview

The final implementation follows an event-driven, serverless architecture in AWS Region ap-south-2. Payroll work is placed on automatic-payroll-queue and consumed by the payroll-worker Lambda. The worker reads the required data from DynamoDB, generates a PDF using the attached ReportLab Lambda Layer, stores the PDF in the private S3 bucket, creates temporary access, sends the notification through SES, and writes the payroll result back to DynamoDB. Failed queue processing can be routed to automatic-payroll-dlq.

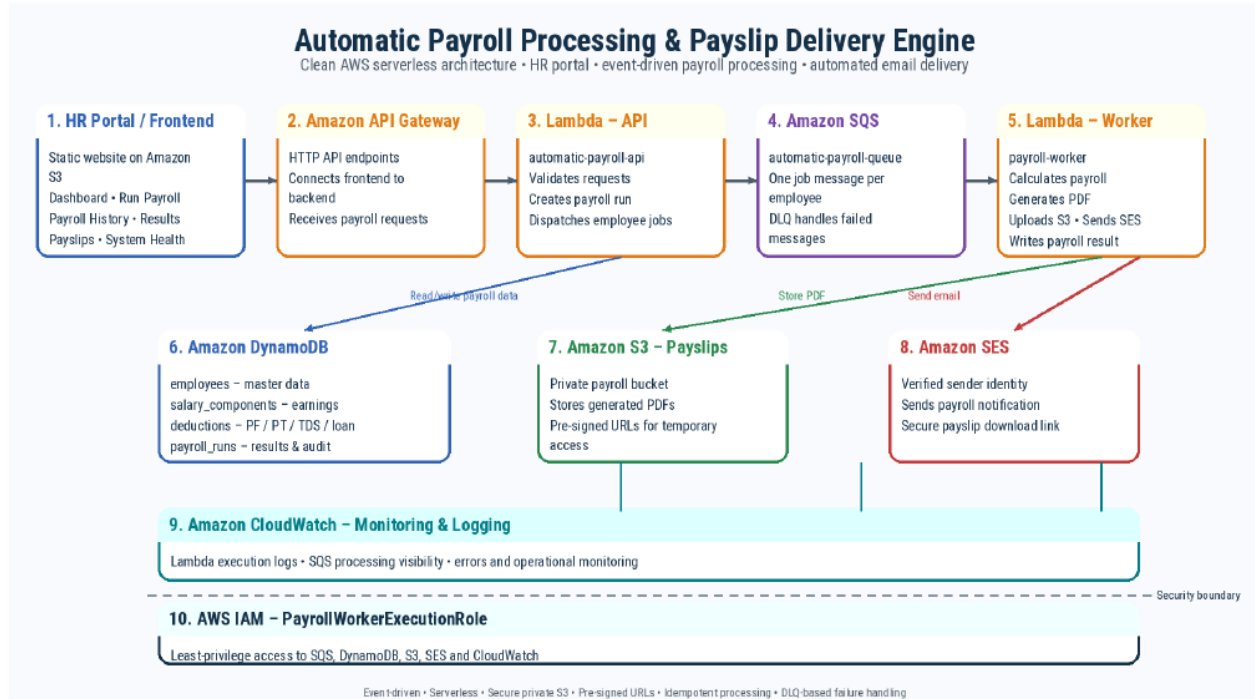


Figure 4.1: Final AWS Architecture Including HR Portal / Frontend

4.2 Architecture Components

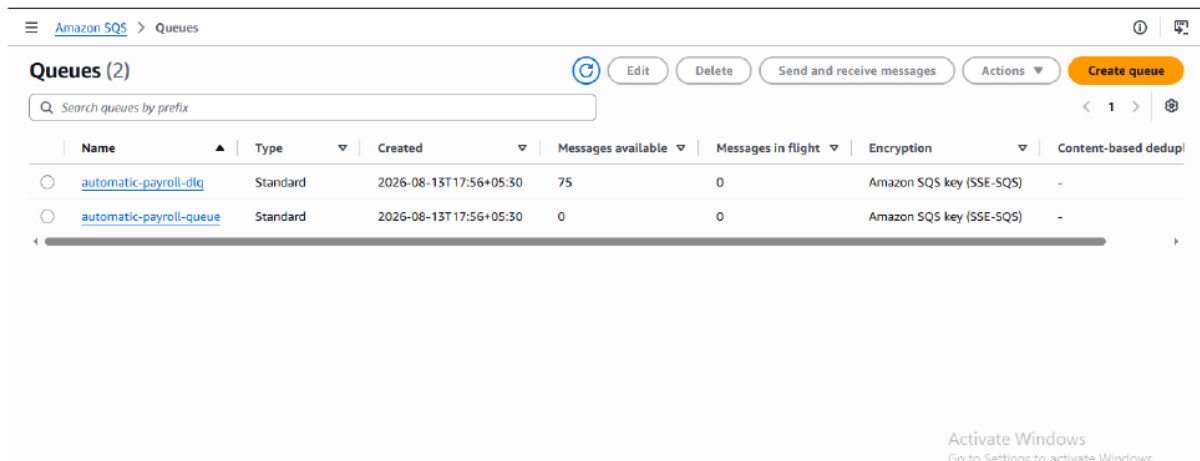
Component	Role
Payroll Request / Trigger	Starts a payroll run and supplies run/month context.
Amazon SQS – automatic-payroll-queue	Buffers employee payroll jobs and decouples producers from workers.
automatic-payroll-dlq	Receives messages that cannot be successfully processed after retries.
AWS Lambda – payroll-worker	Retrieves data, calculates payroll, generates payslip, stores it, sends email, and records the result.
Amazon DynamoDB	Stores employees, salary components, deductions, and payroll runs.
ReportLab Lambda Layer	Provides PDF generation capability to payroll-worker.
Amazon S3 – automatic-payroll-payslips-2026	Private storage for generated payslip PDFs.
Amazon SES	Sends employee payslip notification email.
Amazon CloudWatch	Captures Lambda execution logs and errors.
AWS IAM	Controls Lambda permissions to AWS resources.

4.3 End-to-End Processing Flow

1. Payroll request is created for a payroll month.
2. Employee jobs are placed into automatic-payroll-queue.
3. SQS invokes payroll-worker.
4. Lambda reads employee information from DynamoDB.
5. Lambda reads salary components and deduction parameters.
6. Payroll calculation is performed.
7. ReportLab generates the payslip PDF.
8. The PDF is uploaded to the private S3 bucket.
9. A temporary pre-signed URL is created.
10. SES sends the payslip notification.
11. The payroll result and S3 reference are stored in payroll_runs.
12. CloudWatch records execution details; failed messages can move to the DLQ.

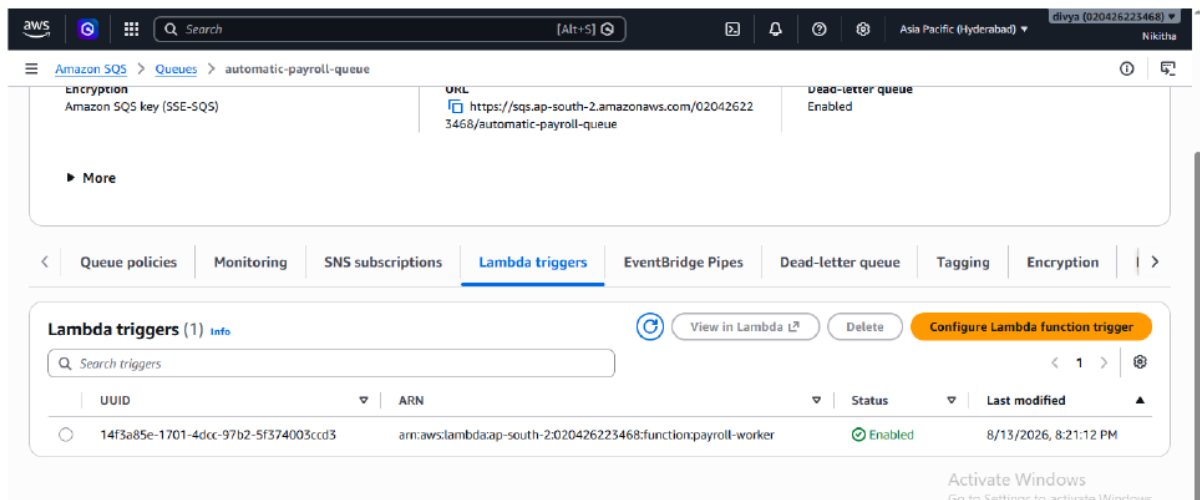
4.4 Queue and Lambda Integration

The submitted AWS configuration confirms that automatic-payroll-queue has an enabled Lambda trigger targeting payroll-worker. The SQS console also shows automatic-payroll-dlq configured for failure handling.



Name	Type	Created	Messages available	Messages in flight	Encryption	Content-based deduplication
automatic-payroll-dlq	Standard	2026-08-13T17:56+05:30	75	0	Amazon SQS key (SSE-SQS)	-
automatic-payroll-queue	Standard	2026-08-13T17:56+05:30	0	0	Amazon SQS key (SSE-SQS)	-

Figure 4.2: Amazon SQS Queues – Payroll Queue and Dead-Letter Queue



UUID	ARN	Status	Last modified
14f3a85e-1701-4dcc-97b2-5f374003ccd3	arn:aws:lambda:ap-south-2:020426223468:function:payroll-worker	Enabled	8/13/2026, 8:21:12 PM

Figure 4.3: SQS Lambda Trigger Configured for payroll-worker

4.5 Lambda Worker

The AWS Lambda console shows payroll-worker deployed with Python 3.14 on x86_64. The code is organized into handler.py, calculator.py, dynamodb_client.py, payroll_service.py, payslip_service.py, and tax_rules.py.

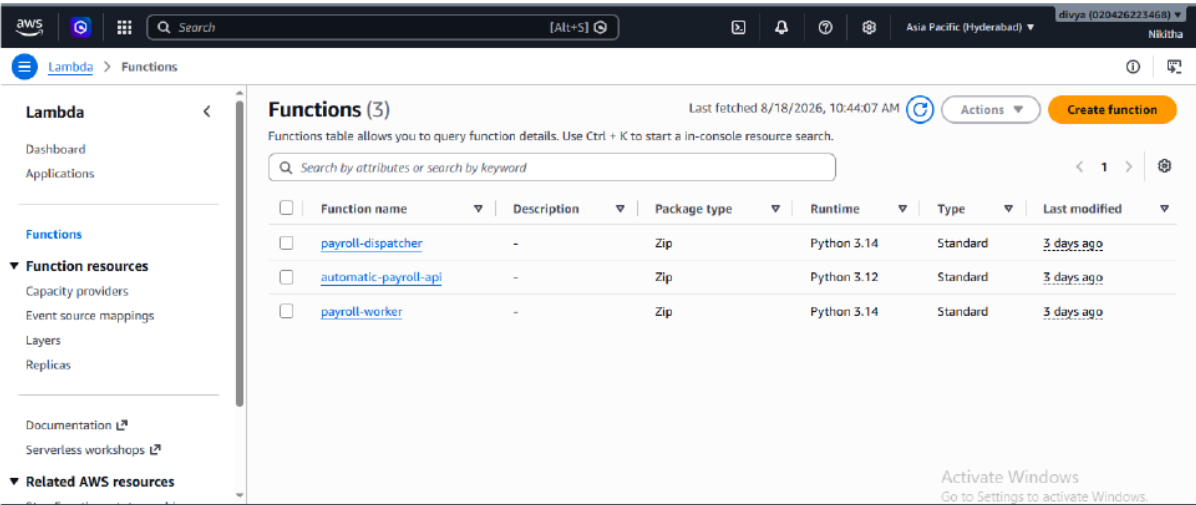


Figure 4.4: Lambda Functions Deployed in the Project

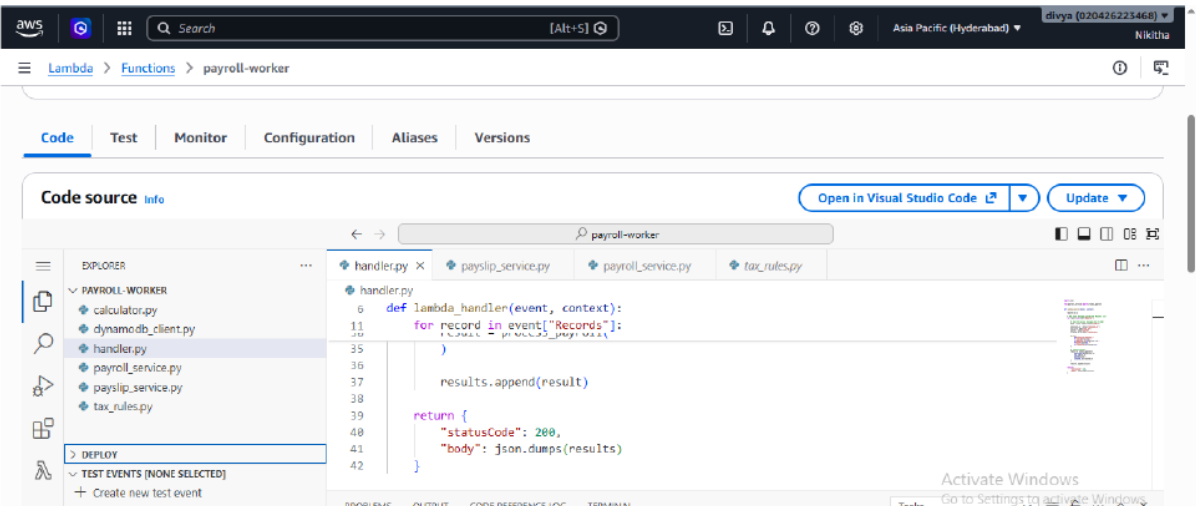


Figure 4.5: payroll-worker Source Structure

4.6 Failure Handling

The queue design includes a dead-letter queue. During testing, CloudWatch also captured an SES MessageRejected error, demonstrating that service errors are observable in the logging layer and can be investigated without losing the surrounding payroll-processing context.

CloudWatch > Log management > /aws/lambda/payroll-worker > 2026/08/18/[\$LATEST]abb7326c2f844c3481c1d5bba802ef37

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search 1m 1h UTC timezone Display

Timestamp	Message
2026-08-18T04:55:48.625Z	START RequestId: 616fc61c-8491-5c79-af4f-a4e16367b2bb Version: \$LATEST
2026-08-18T04:55:48.625Z	Processing payroll: run_id=RUN-2026-08-1b913b99, employee_id=EMP002, month=2026-08, created_at=2026-08-18.
2026-08-18T04:55:48.659Z	Generating payslip for EMP002
2026-08-18T04:55:48.761Z	Payslip uploaded: s3://automatic-payroll-payslips-2026/payslips/2026/08/RUN-2026-08-1b913b99/EMP002.pdf
2026-08-18T04:55:48.762Z	Pre-signed URL generated for payslips/2026/08/RUN-2026-08-1b913b99/EMP002.pdf
2026-08-18T04:55:48.825Z	[ERROR] MessageRejected: An error occurred (MessageRejected) when calling the SendEmail operation: Email ..
2026-08-18T04:55:48.827Z	END RequestId: 616fc61c-8491-5c79-af4f-a4e16367b2bb
2026-08-18T04:55:48.827Z	REPORT RequestId: 616fc61c-8491-5c79-af4f-a4e16367b2bb Duration: 281.30 ms Billed Duration: 282 ms Memory: ..

No newer events at this moment. Auto retry paused. [Resume](#)

Activate Windows. Go to Settings to activate Windows. [Back to top](#)

Figure 4.6: CloudWatch Payroll Processing Log with SES Error Evidence

Chapter 5 – Database Design

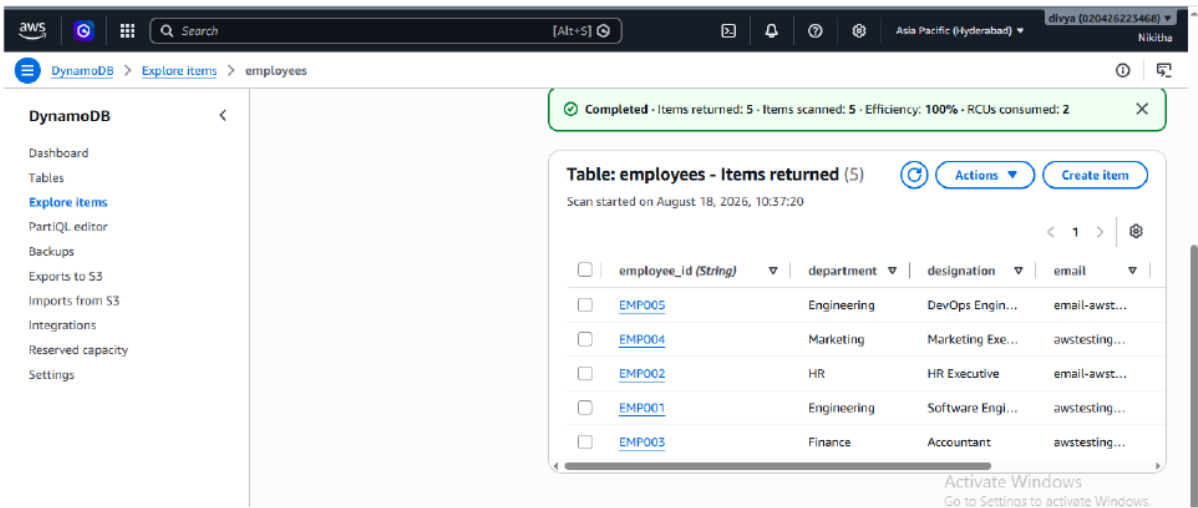
5.1 Database Overview

Amazon DynamoDB is used as the primary database. The screenshots supplied for the completed project show four logical tables: employees, salary_components, deductions, and payroll_runs.

Table	Purpose	Key
employees	Employee master data, department, designation and contact information	employee_id
salary_components	Basic salary and allowance inputs	employee_id
deductions	PF, TDS and loan deduction inputs	employee_id
payroll_runs	Calculated payroll results and payslip information	run_id + employee_id

5.2 employees Table

The employees table contains five sample employee records. The screenshot shows employee IDs EMP001 through EMP005 along with department, designation, and email fields.



The screenshot displays the AWS DynamoDB console interface for the 'employees' table. A green status bar at the top indicates 'Completed' with 5 items returned, 5 items scanned, 100% efficiency, and 2 RCUs consumed. The table view shows 5 items with columns: employee_id (String), department, designation, and email. The items are listed as follows:

employee_id (String)	department	designation	email
EMP005	Engineering	DevOps Engin...	email-awst...
EMP004	Marketing	Marketing Exe...	awstesting...
EMP002	HR	HR Executive	email-awst...
EMP001	Engineering	Software Engi...	awstesting...
EMP003	Finance	Accountant	awstesting...

Figure 5.1: DynamoDB employees Table

5.3 salary_components Table

The salary_components table stores base salary, HRA, other allowances, and transport allowance. The supplied records include EMP001 through EMP005.

Table: salary_components - Items returned (5)

Scan started on August 18, 2025, 10:38:37

employee_id	base_salary	hra	other_allowances	transportation_allowance
EMP001	40000	12000	2000	3000
EMP002	45000	14000	2500	3000
EMP003	35000	10000	1500	2500
EMP004	30000	9000	1000	2000
EMP005	55000	18000	3000	4000

Figure 5.2: DynamoDB salary_components Table

5.4 deductions Table

The deductions table contains loan deduction, PF percentage, and TDS rate for each employee. The submitted data shows PF percentage values of 12 and employee-specific TDS and loan deductions.

Table: deductions - Items returned (5)

Scan started on August 18, 2025, 10:39:43

employee_id	loan_deduction	pf_percentage	tds_rate
EMP005	5000	12	10
EMP004	1000	12	5
EMP002	0	12	10
EMP001	3000	12	10
EMP003	2000	12	8

Figure 5.3: DynamoDB deductions Table

5.5 payroll_runs Table

The payroll_runs table contains the results of processed payroll jobs. The submitted screenshot shows 50 returned items and includes run IDs, employee IDs, base salary, creation information, deduction totals, and email-related information.

run_id	employee_id	basic_salary	created_date	deductions	email
RUN-2026-...	EMP001	40000	2026-08-...	13750	01200
RUN-2026-...	EMP003	35000	2026-08-...	10320	01200
RUN-2026-...	EMP004	30000	2026-08-...	6900	01200
RUN-2025-...	EMP001	40000	2026-08-...	13750	01200
RUN-2025-...	EMP003	35000	2026-08-...	10320	01200
RUN-2025-...	EMP004	30000	2026-08-...	6900	01200
RUN-2025-...	EMP001	40000	2026-08-...	13750	01200

Figure 5.4: DynamoDB payroll_runs Table

5.6 Payroll Calculation Model

Calculation	Formula
Gross Salary	Basic Salary + HRA + Transport Allowance + Other Allowances
PF	$\text{Basic Salary} \times \text{PF Percentage} / 100$
Professional Tax	Simplified slab-based rule
TDS	$\text{Gross Salary} \times \text{TDS Rate} / 100$
Total Deductions	PF + Professional Tax + TDS + Loan Deduction
Net Pay	Gross Salary – Total Deductions

5.7 Sample Calculation from Payslip

Component	EMP004 – August 2026
Basic Salary	Rs. 30,000.00
HRA	Rs. 9,000.00
Transport Allowance	Rs. 2,000.00
Other Allowances	Rs. 1,000.00
Gross Salary	Rs. 42,000.00
PF	Rs. 3,600.00
Professional Tax	Rs. 200.00
TDS	Rs. 2,100.00
Loan Deduction	Rs. 1,000.00
Total Deductions	Rs. 6,900.00
Net Pay	Rs. 35,100.00

The above values are directly visible in the submitted EMP004 payslip and illustrate the end-to-end calculation result. The tax rules used by the demonstration should not be interpreted as a complete statutory payroll implementation.

Chapter 6 – System Implementation

6.1 Lambda Implementation

The payroll-worker function is implemented in Python 3.14 and uses a modular source structure. The handler receives SQS records and invokes the payroll processing service.

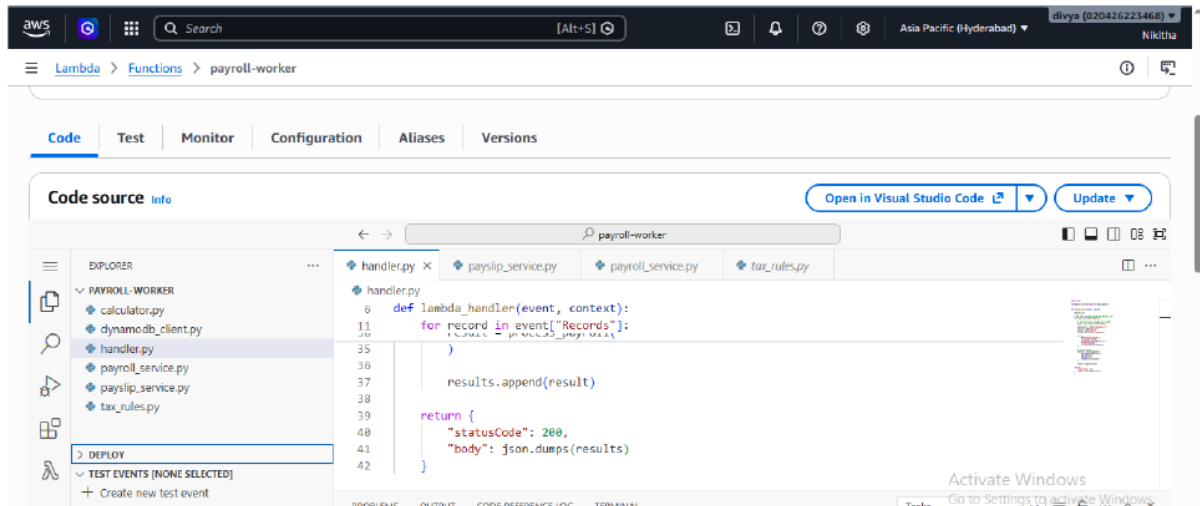


Figure 6.1: payroll-worker Code and Module Structure

6.2 ReportLab Lambda Layer

The payroll-worker has one Lambda Layer attached. The supplied AWS screenshot identifies the layer as payroll-reportlab-layer, version 1, compatible with Python 3.14 and x86_64. This layer provides the ReportLab dependency required for PDF generation.

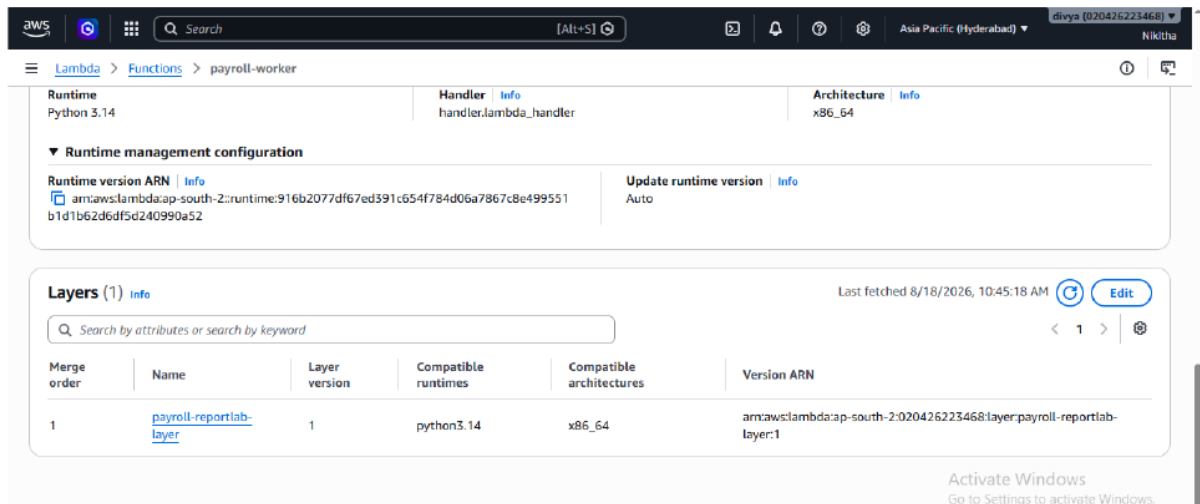


Figure 6.2: payroll-reportlab-layer Attached to payroll-worker

6.3 Environment Configuration

The Lambda configuration contains an SES_SENDER_EMAIL environment variable. The supplied screenshot confirms that the sender email is configured in the function environment. In a production deployment, secrets and sensitive configuration should be managed using appropriate secret-management practices.

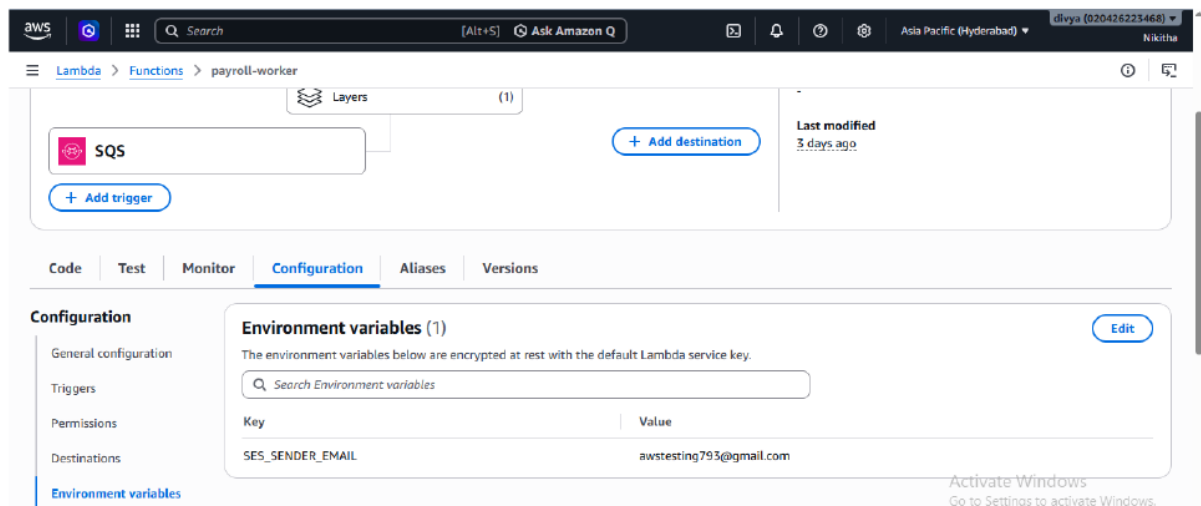


Figure 6.3: payroll-worker Environment Variable

6.4 PDF Payslip Generation

The system generates a formatted salary slip containing employer information, employee name and ID, payroll month, payroll status, earnings, deductions, total deductions, and net pay. The submitted payslip demonstrates a successful calculated result for EMP004 for August 2026.

NIKITA LOGISTICS PVT. LTD.

SALARY SLIP

Plot No. 42, Sector 19, Logistics Park, Vashi, Navi Mumbai, MH - 400703
CIN: U63090MH2018PTC312890 - GSTIN: 27AAACN8821F1Z4 - hr@nikitalogistics.com

Employee Name	Sneha Reddy	Employee ID	EMP004
Month	2026-08	Payroll Status	CALCULATED

EARNINGS

Basic Salary	Rs. 30,000.00
HRA	Rs. 9,000.00
Transport Allowance	Rs. 2,000.00
Other Allowances	Rs. 1,000.00
Gross Salary	Rs. 42,000.00

DEDUCTIONS

PF	Rs. 3,600.00
Professional Tax	Rs. 200.00
TDS	Rs. 2,100.00
Loan Deduction	Rs. 1,000.00
Total Deductions	Rs. 6,900.00

NET PAY	Rs. 35,100.00
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Nikita Logistics Pvt. Ltd. — Payroll & HR Operations
This is a computer-generated salary slip. No physical signature is required.
Processed via AWS Serverless Automated Payroll Engine

Figure 6.4: Generated PDF Payslip – EMP004, August 2026

6.5 S3 Payslip Storage

The submitted S3 screenshot shows EMP001.pdf stored in the automatic-payroll-payslips-2026 bucket in the ap-south-2 region. The object path follows a payroll-oriented folder structure containing year, month, run ID, and employee ID.

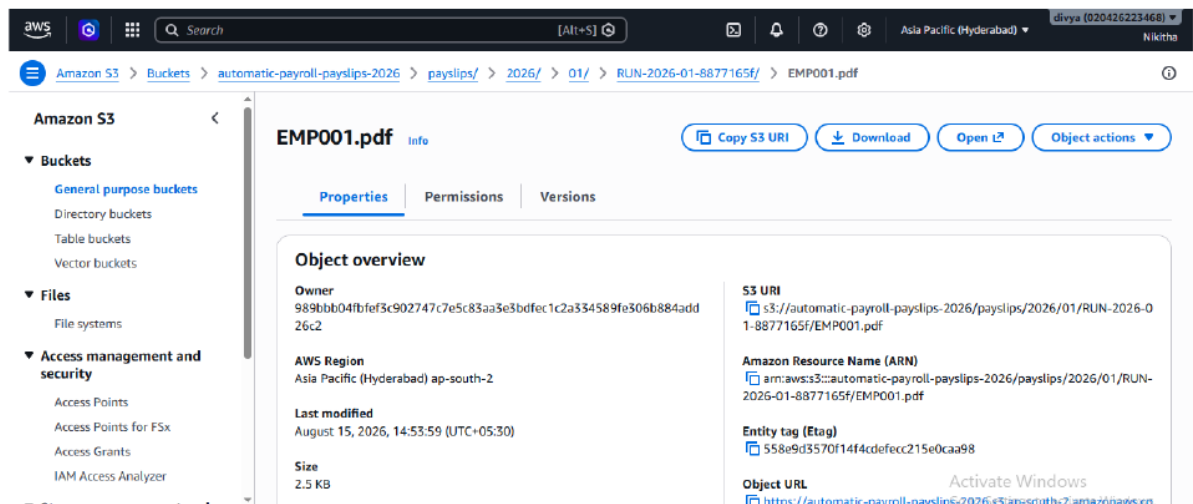


Figure 6.5: Generated Payslip Stored in Amazon S3

6.6 Pre-Signed URL and Email Delivery

After the payslip is stored, the system creates temporary access to the private document and includes the access link in the employee email. The supplied email screenshot shows the completed payslip notification and a “View/Download Payslip” action.

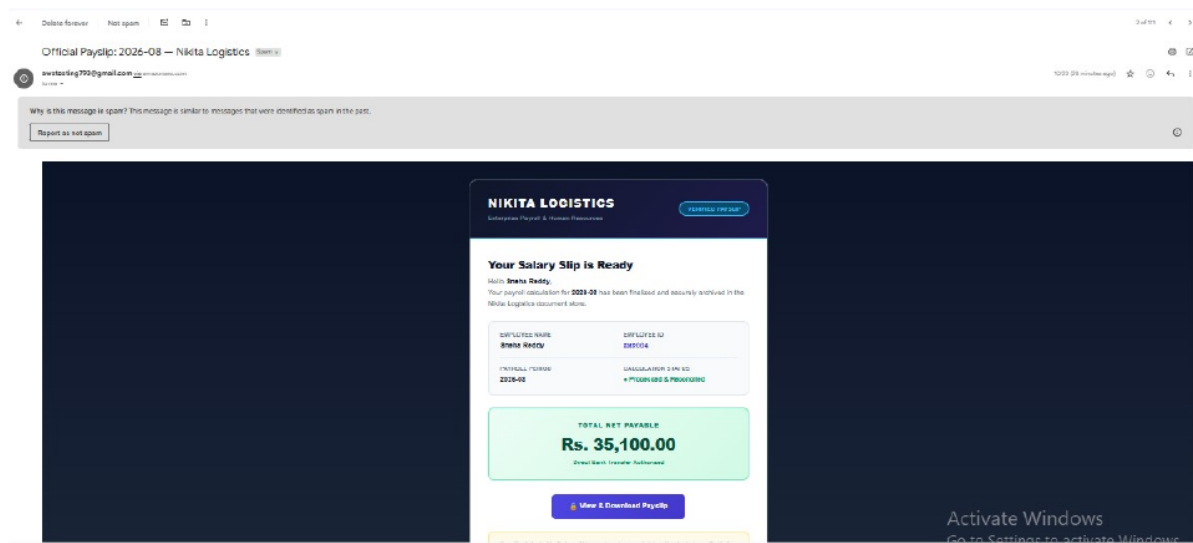


Figure 6.6: Payslip Email Delivered to Employee

6.7 IAM Security

The PayrollWorkerExecutionRole is configured with AWS-managed Lambda execution and SQS execution policies, together with customer-managed DynamoDB and S3 policies and an SES access policy. This demonstrates resource-specific permissions rather than relying on unrestricted access.

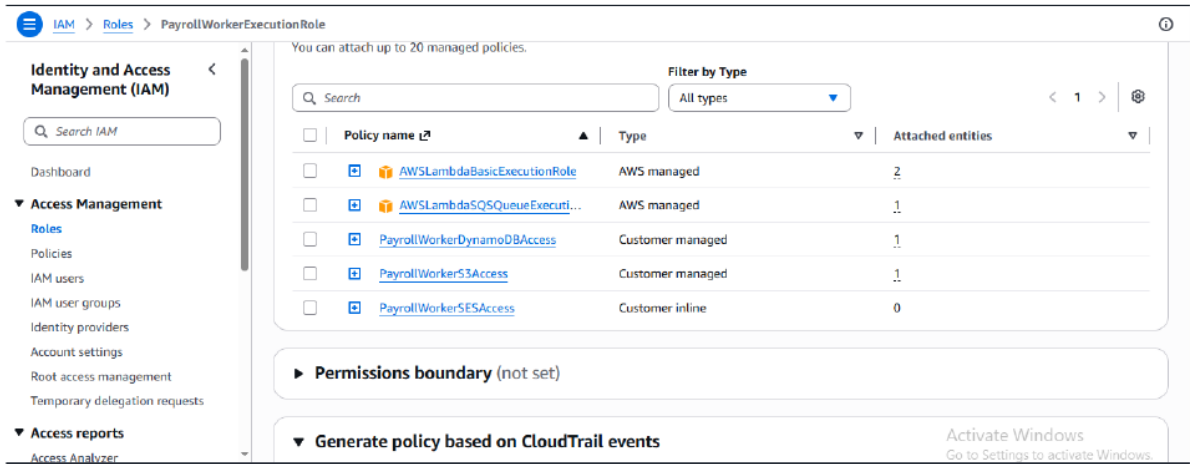


Figure 6.7: PayrollWorkerExecutionRole IAM Policies

6.8 CloudWatch Monitoring

CloudWatch is used to observe the payroll-worker execution. The supplied log shows the request starting, payroll processing for a run and employee, payslip generation, S3 upload, pre-signed URL generation, and an SES MessageRejected error. The logging trail therefore provides useful operational evidence for troubleshooting the email stage.

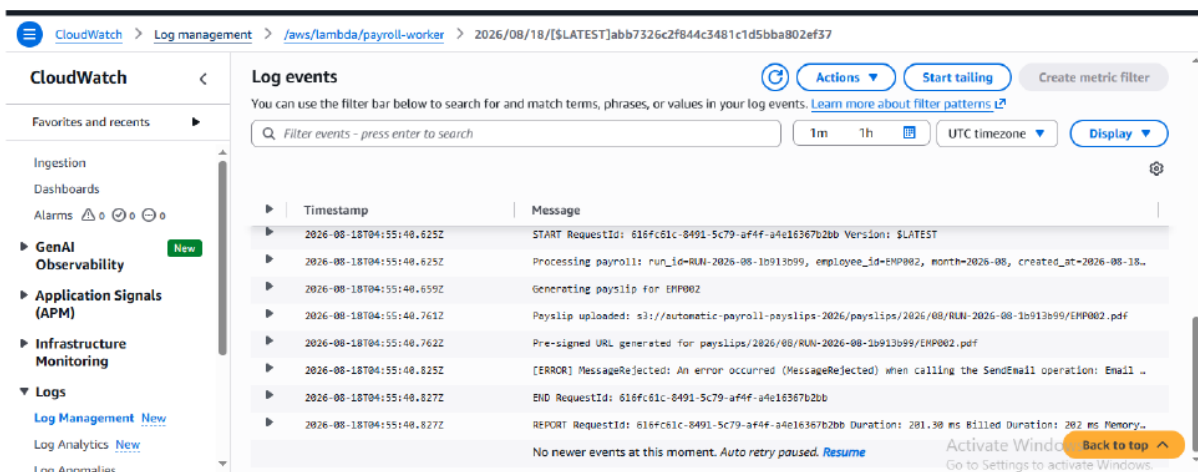
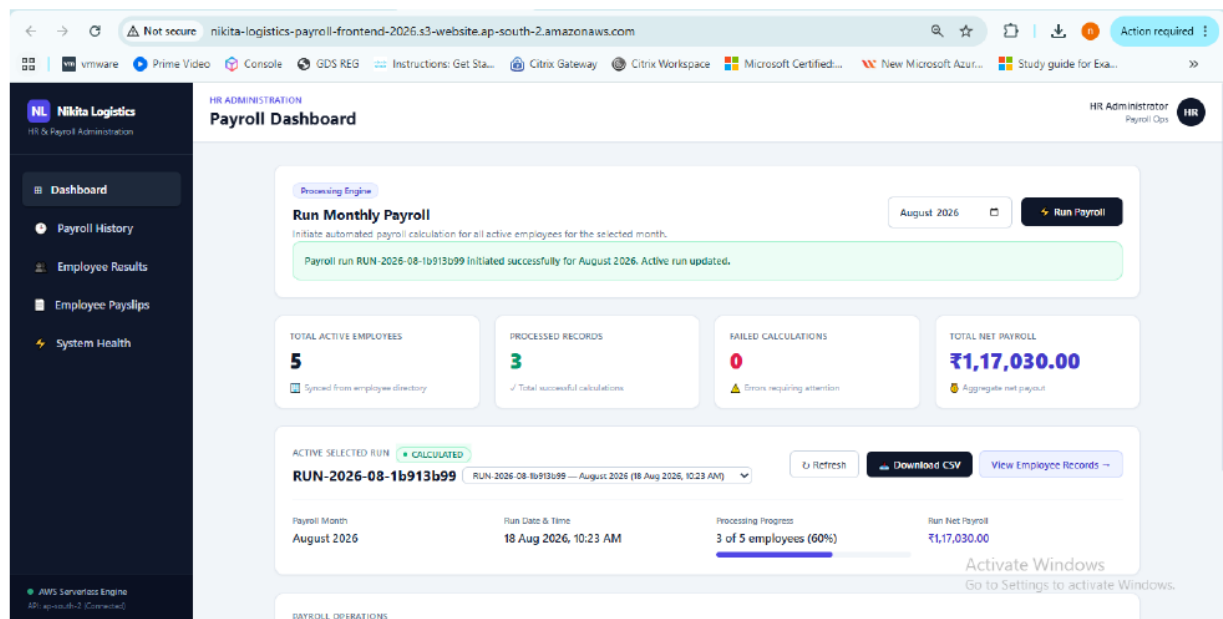


Figure 6.8: CloudWatch Execution Log

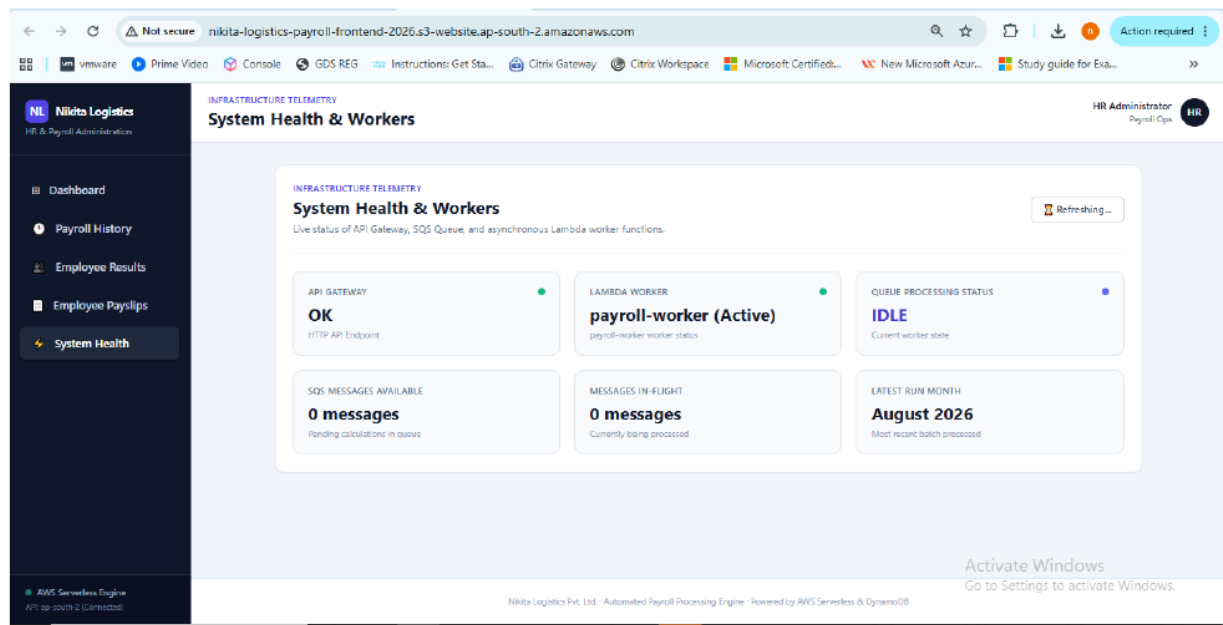
6.9.6 Dashboard Summary

Figure 6.13: HR Portal – Dashboard



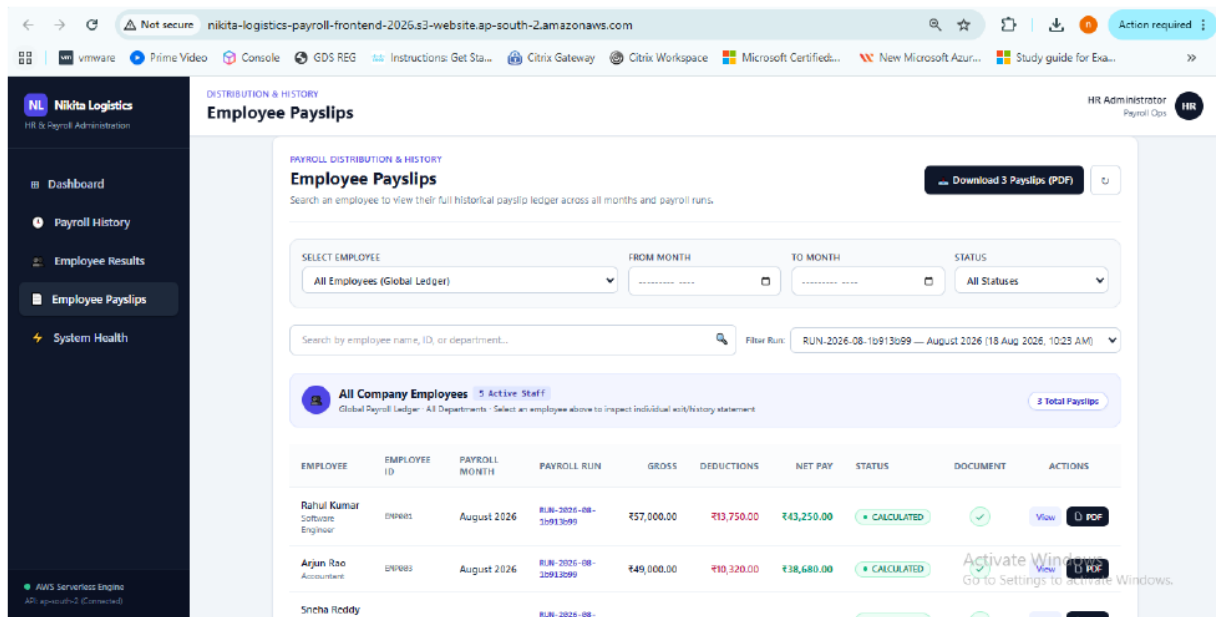
6.9.5 System Health and Workers

Figure 6.12: System Health & Workers



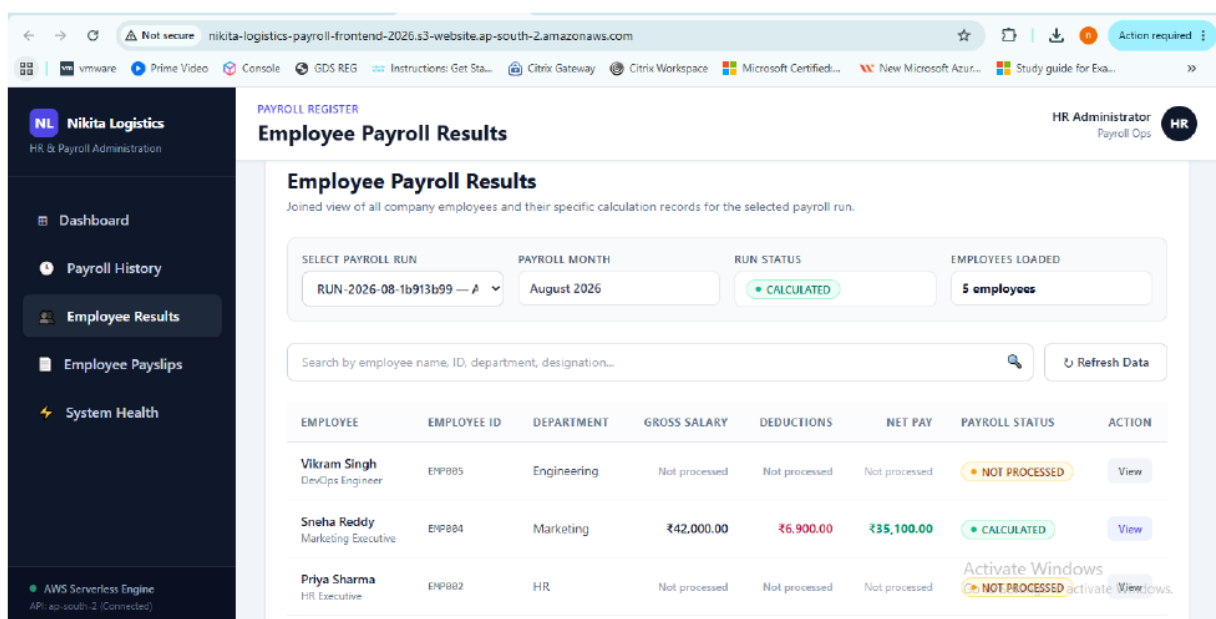
6.9.4 Employee Payslips

Figure 6.11: Employee Payslips



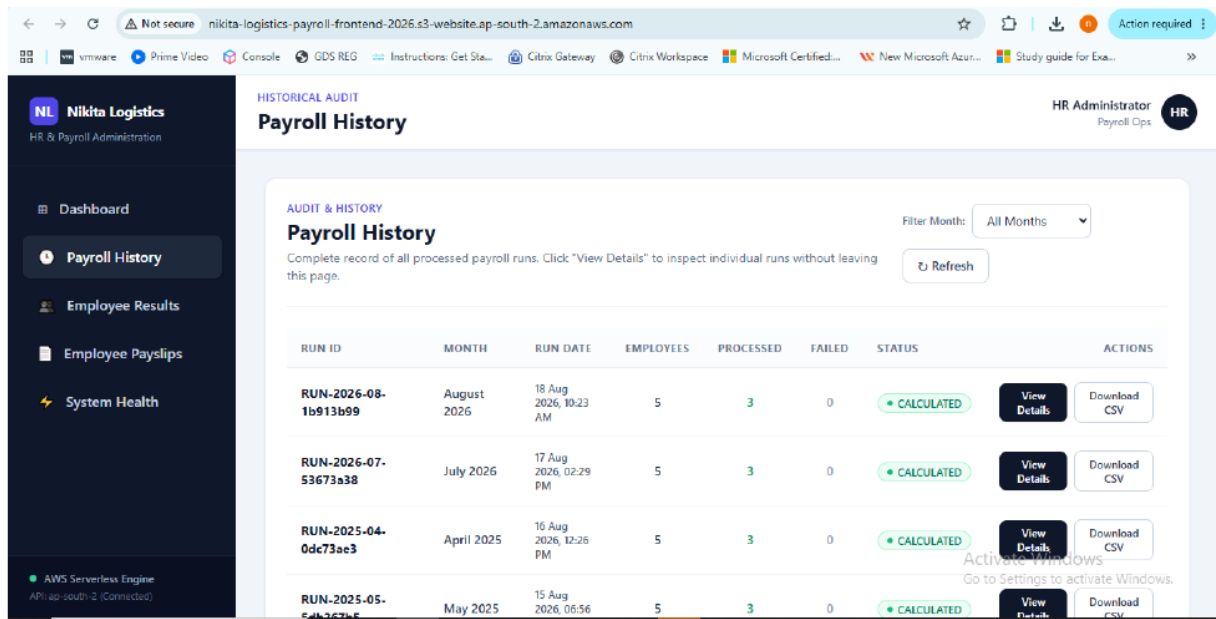
6.9.3 Employee Payroll Results

Figure 6.10: Employee Payroll Results



6.9.2 Payroll History

Figure 6.9: Payroll History



The HR portal is a static web frontend hosted on Amazon S3. It provides an administrative interface for initiating payroll runs and reviewing payroll history, employee-level results, payslip records, and system health. The frontend communicates with the backend through HTTP API endpoints and presents current payroll state to the HR administrator.

6.9.1 HR Dashboard

The dashboard provides the primary HR administration view. The administrator can select a payroll month and initiate a payroll run. The supplied screenshot shows the active run, active employee count, processed records, failed calculations, total net payroll, and processing progress.

6.9 Frontend and HR Portal

Chapter 7 – Testing and Results

7.1 Testing Approach

Testing was performed across the major components of the serverless payroll pipeline. Evidence supplied by the completed project includes populated DynamoDB tables, an active SQS-to-Lambda trigger, deployed Lambda functions and layer, generated S3 payslip objects, a generated PDF payslip, an employee email notification, and CloudWatch execution logs.

7.2 Test Environment

Parameter	Observed Configuration
AWS Region	Asia Pacific (Hyderabad) – ap-south-2
Payroll Worker Runtime	Python 3.14
Architecture	x86_64
Queue	automatic-payroll-queue
Dead-Letter Queue	automatic-payroll-dlq
Database	Amazon DynamoDB
Object Storage	Amazon S3
Email Service	Amazon SES

Monitoring	Amazon CloudWatch
PDF Dependency	payroll-reportlab-layer

7.3 Test Cases

ID	Test Case	Evidence	Result
TC-01	Employee data availability	employees table screenshot	Successful
TC-02	Salary component retrieval	salary_components table screenshot	Successful
TC-03	Deduction retrieval	deductions table screenshot	Successful
TC-04	Payroll result persistence	payroll_runs table screenshot	Successful
TC-05	SQS queue availability	SQS console screenshot	Successful
TC-06	SQS → Lambda integration	Lambda trigger screenshot	Successful
TC-07	Lambda deployment	Lambda console screenshot	Successful
TC-08	PDF dependency	ReportLab layer screenshot	Successful
TC-09	PDF payslip generation	EMP004 payslip screenshot	Successful
TC-10	S3 payslip storage	EMP001.pdf S3 object screenshot	Successful
TC-11	Employee email delivery	Received payslip email screenshot	Successful
TC-12	Operational logging	CloudWatch screenshot	Logged; SES error observed in one execution

7.4 End-to-End Result

The supplied evidence demonstrates that the core payroll pipeline was implemented and exercised: payroll data exists in DynamoDB, jobs are connected from SQS to Lambda, the worker is deployed with its ReportLab layer, payslips are generated as PDFs, objects are stored in S3, and an employee-facing payslip email is produced. CloudWatch provides the corresponding execution trail.

7.5 Observed Error Handling

One submitted CloudWatch execution shows an SES MessageRejected error after successful payroll processing, payslip generation, S3 upload, and pre-signed URL generation. This is useful evidence that the system logs failures at the email stage and can be diagnosed from the execution trail. The separately supplied email screenshot demonstrates a successfully delivered payslip notification, indicating that email delivery was also successfully exercised.

7.6 Security and Reliability Observations

- S3 is used as the document store rather than exposing the PDF through the application process.
- Temporary access is provided through a pre-signed link.
- IAM policies are attached to a dedicated PayrollWorkerExecutionRole.
- SQS provides queue-based decoupling and retry behavior.
- A dead-letter queue is configured for failed messages.
- CloudWatch provides execution visibility.

Chapter 8 – Conclusion and Future Scope

8.1 Conclusion

The Automatic Payroll Processing and Payslip Delivery Engine demonstrates a practical serverless approach to automating payroll operations. The final implementation combines Amazon SQS, AWS Lambda, Amazon DynamoDB, Amazon S3, Amazon SES, CloudWatch, IAM, and a ReportLab Lambda Layer.

The submitted evidence confirms the presence of employee and payroll data, queue-to-Lambda integration, modular Lambda implementation, PDF generation, S3 storage, employee email delivery, IAM controls, and CloudWatch logging. The project therefore provides an end-to-end cloud-native foundation for automated payroll and payslip delivery.

8.2 Project Achievements

- Implemented queue-driven payroll processing.
- Implemented modular Python Lambda worker.
- Implemented DynamoDB-based payroll data storage.
- Implemented PDF payslip generation using ReportLab.
- Stored payslips in Amazon S3.
- Implemented temporary payslip access.
- Integrated Amazon SES for employee notifications.
- Configured a dedicated IAM execution role.
- Configured a dead-letter queue.
- Implemented CloudWatch operational logging.

8.3 Learning Outcomes

- AWS serverless architecture
- Amazon SQS and event-driven processing
- AWS Lambda with Python
- DynamoDB data modeling
- S3 object storage and document access
- SES integration
- IAM least-privilege concepts
- CloudWatch logging and troubleshooting
- Lambda Layers and dependency packaging
- PDF generation with ReportLab
- Cloud-based project testing and documentation

8.4 Limitations

- The payroll tax rules are simplified for demonstration purposes.
- Production statutory compliance has not been implemented.
- Sample employee data is used for the demonstration.
- SES delivery can be affected by account verification/sandbox restrictions.
- Attendance, leave, benefits, and HRMS integrations are outside the current scope.
- Production deployments would require stronger secret management, governance, monitoring, and compliance controls.

8.5 Future Scope

- Integration with attendance and leave systems.

- Integration with a production HRMS.
- Organization-specific statutory tax and payroll rules.
- Payroll approval and finalization workflow.
- HR/admin dashboard for payroll runs.
- CSV/Excel payroll register export.
- Scheduled monthly payroll execution.
- Enhanced SES delivery and bounce monitoring.
- Centralized audit reports.
- Enterprise key management and encryption controls.
- Multi-tenant payroll processing.
- Automated reconciliation and validation.

8.6 Final Remarks

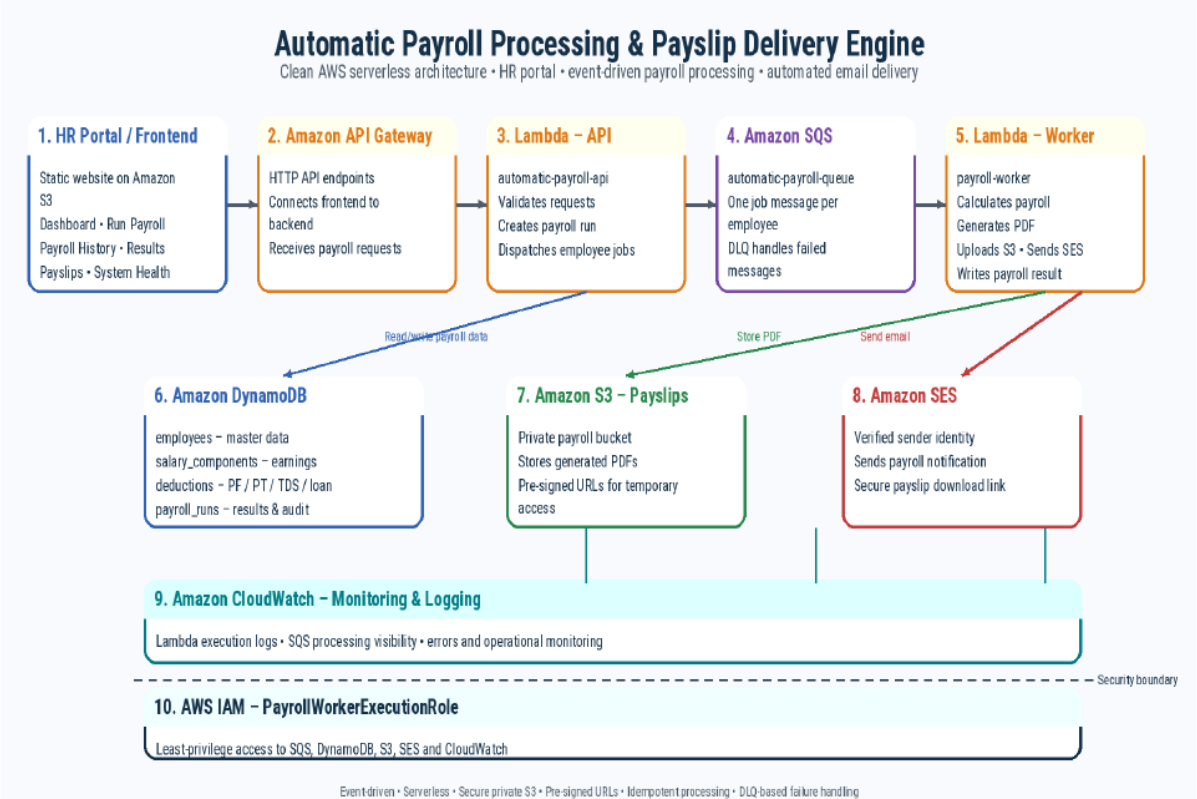
The project provided practical experience in designing, implementing, testing, troubleshooting, and documenting a cloud-native enterprise workflow. It demonstrates how managed AWS services can replace repetitive payroll activities with an event-driven, scalable, auditable, and secure architecture.

References

1. Amazon Web Services – AWS Lambda Developer Guide.
2. Amazon Web Services – Amazon SQS Developer Guide.
3. Amazon Web Services – Amazon DynamoDB Developer Guide.
4. Amazon Web Services – Amazon S3 User Guide.
5. Amazon Web Services – Amazon SES Developer Guide.
6. Amazon Web Services – Amazon CloudWatch User Guide.
7. Amazon Web Services – AWS Identity and Access Management User Guide.
8. ReportLab Documentation.

Appendices

Appendix A – Architecture Diagram



Appendix Figure A.1: Final AWS Architecture

Appendix B – Database Evidence

Table: employees - Items returned (5)

Scan started on August 18, 2026, 10:37:20

employee_id (String)	department	designation	email
EMP005	Engineering	DevOps Engin...	email-awst...
EMP004	Marketing	Marketing Exe...	awstesting...
EMP002	HR	HR Executive	email-awst...
EMP001	Engineering	Software Engi...	awstesting...
EMP003	Finance	Accountant	awstesting...

Appendix Figure B.1: employees

DynamoDB > Explore items > salary_components

Completed - Items returned: 5 - Items scanned: 5 - Efficiency: 100% - RCUs consumed: 4

Table: salary_components - Items returned (5)

Scan started on August 18, 2026, 10:38:37

	employ...	base_salary	hra	other_a...	transpo...
☐	EMP001	40000	12000	2000	3000
☐	EMP002	45000	14000	2500	3000
☐	EMP003	35000	10000	1500	2500
☐	EMP004	30000	9000	1000	2000
☐	EMP005	55000	18000	3000	4000

Activate Windows
Go to Settings to activate Windows.

Appendix Figure B.2: salary_components

DynamoDB > Explore items > deductions

Completed - Items returned: 5 - Items scanned: 5 - Efficiency: 100% - RCUs consumed: 2

Table: deductions - Items returned (5)

Scan started on August 18, 2026, 10:39:43

	employ...	loan_deduction	pf_percentage	tds_rate
☐	EMP005	5000	12	10
☐	EMP004	1000	12	5
☐	EMP002	0	12	10
☐	EMP001	3000	12	10
☐	EMP003	2000	12	8

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Appendix Figure B.3: deductions

DynamoDB > Explore items > payroll_runs

Table: payroll_runs - Items returned (50)

Scan started on August 18, 2026, 10:40:05

	run_id (...)	employ...	base_s...	created...	deducti...	email...
☐	RUN-2026-...	EMP001	40000	2026-08-...	13750	01200...
☐	RUN-2026-...	EMP003	35000	2026-08-...	10320	01200...
☐	RUN-2026-...	EMP004	30000	2026-08-...	6900	01200...
☐	RUN-2025-...	EMP001	40000	2026-08-...	13750	01200...
☐	RUN-2025-...	EMP003	35000	2026-08-...	10320	01200...
☐	RUN-2025-...	EMP004	30000	2026-08-...	6900	01200...
☐	RUN-2025-...	EMP001	40000	2026-08-...	13750	01200...

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Appendix Figure B.4: payroll_runs

Appendix C – AWS Integration Evidence

Amazon SQS > Queues

Queues (2)

Search queues by prefix

Name	Type	Created	Messages available	Messages in flight	Encryption	Content-based dedup
automatic-payroll-dlg	Standard	2026-08-13T17:56+05:30	75	0	Amazon SQS key (SSE-SQS)	-
automatic-payroll-queue	Standard	2026-08-13T17:56+05:30	0	0	Amazon SQS key (SSE-SQS)	-

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Appendix Figure C.1: SQS Queues

Amazon SQS > Queues > automatic-payroll-queue

Encryption: Amazon SQS key (SSE-SQS)

URL: <https://sqs.ap-south-2.amazonaws.com/020426223468/automatic-payroll-queue>

Dead-letter queue: Enabled

Queue policies | Monitoring | SNS subscriptions | **Lambda triggers** | EventBridge Pipes | Dead-letter queue | Tagging | Encryption

Lambda triggers (1)

Search triggers

UUID	ARN	Status	Last modified
14f3a85e-1701-4dcc-97b2-5f374003ccd3	arn:aws:lambda:ap-south-2:020426223468:function:payroll-worker	Enabled	8/13/2026, 8:21:12 PM

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Appendix Figure C.2: Lambda Trigger

Lambda > Functions

Functions (3)

Last fetched 8/18/2026, 10:44:07 AM

Search by attributes or search by keyword

Function name	Description	Package type	Runtime	Type	Last modified
payroll-dispatcher	-	Zip	Python 3.14	Standard	3 days ago
automatic-payroll-api	-	Zip	Python 3.12	Standard	3 days ago
payroll-worker	-	Zip	Python 3.14	Standard	3 days ago

Activate Windows
Go to Settings to activate Windows.

Appendix Figure C.3: Lambda Functions

divya (020426223468) Nikitha

Search [Alt+S]

Asia Pacific (Hyderabad)

Lambda > Functions > payroll-worker

Runtime
Python 3.14

Handler
handler.lambda_handler

Architecture
x86_64

Runtime management configuration

Runtime version ARN
arn:aws:lambda:ap-south-2::runtime:916b2077df67ed391c654f784d06a7867c8e499551b1d1b62d6df5d240990a52

Update runtime version
Auto

Layers (1)

Last fetched 8/18/2026, 10:45:18 AM

Edit

Search by attributes or search by keyword

< 1 >

Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN
1	payroll-reportlab-layer	1	python3.14	x86_64	arn:aws:lambda:ap-south-2:020426223468:layer:payroll-reportlab-layer:1

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Appendix Figure C.4: ReportLab Layer

Appendix D – Payslip and Delivery Evidence

NIKITA LOGISTICS PVT. LTD.

SALARY SLIP

Plot No. 42, Sector 19, Logistics Park, Vashi, Navi Mumbai, MH – 400703
CIN: U63090MH2018PTC312890 · GSTIN: 27AAACN8821F1Z4 · hr@nikitalogistics.com

Employee Name	Sneha Reddy	Employee ID	EMP004
Month	2026-08	Payroll Status	CALCULATED

EARNINGS

Basic Salary	Rs. 30,000.00
HRA	Rs. 9,000.00
Transport Allowance	Rs. 2,000.00
Other Allowances	Rs. 1,000.00
Gross Salary	Rs. 42,000.00

DEDUCTIONS

PF	Rs. 3,600.00
Professional Tax	Rs. 200.00
TDS	Rs. 2,100.00
Loan Deduction	Rs. 1,000.00
Total Deductions	Rs. 6,900.00

NET PAY

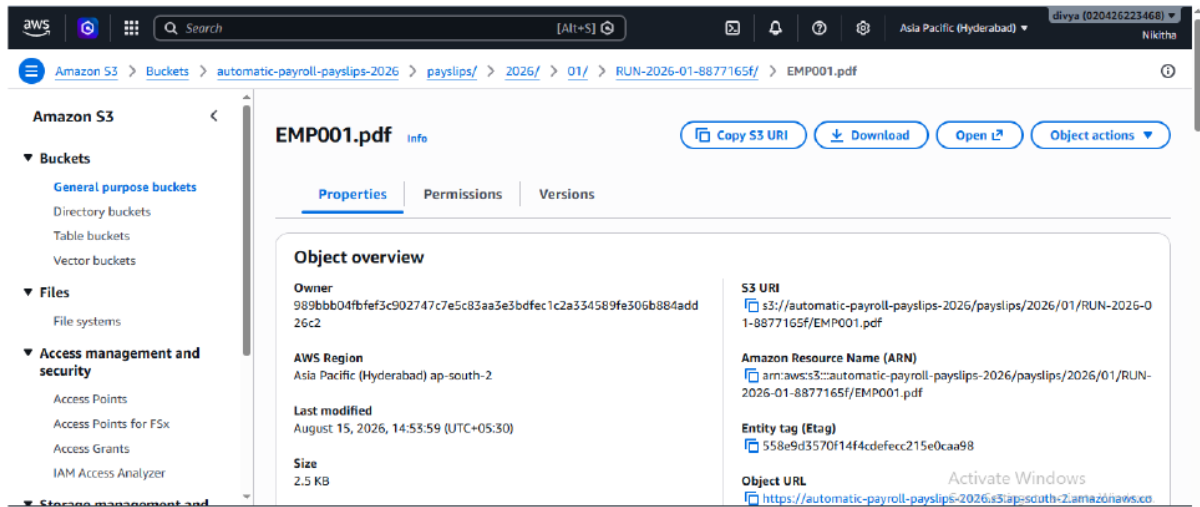
Rs. 35,100.00

Nikita Logistics Pvt. Ltd. — Payroll & HR Operations

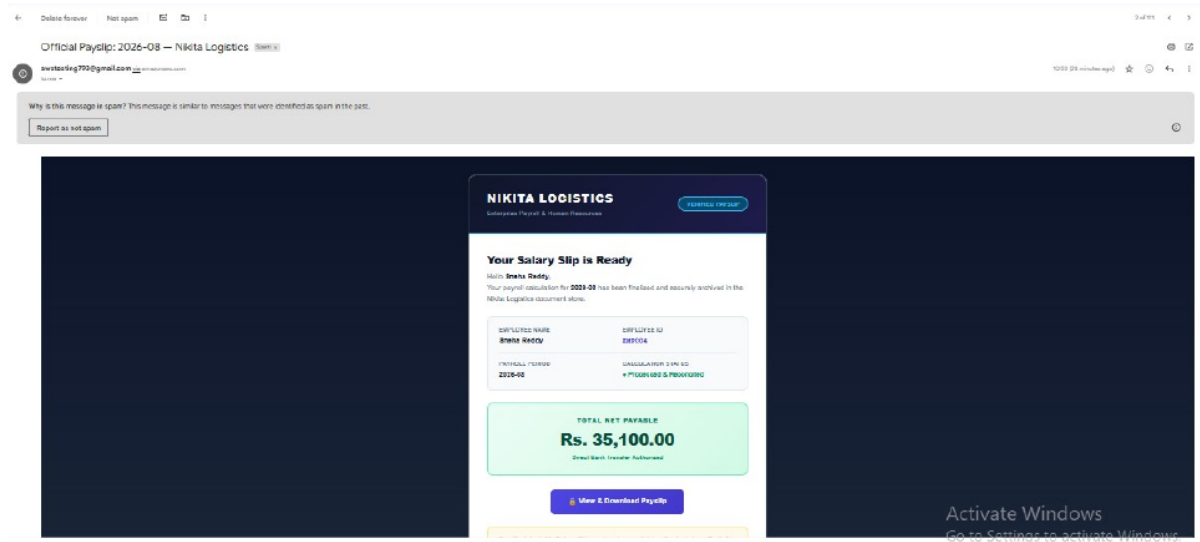
This is a computer-generated salary slip. No physical signature is required.

Processed via AWS Serverless Automated Payroll Engine

Appendix Figure D.1: Generated Payslip

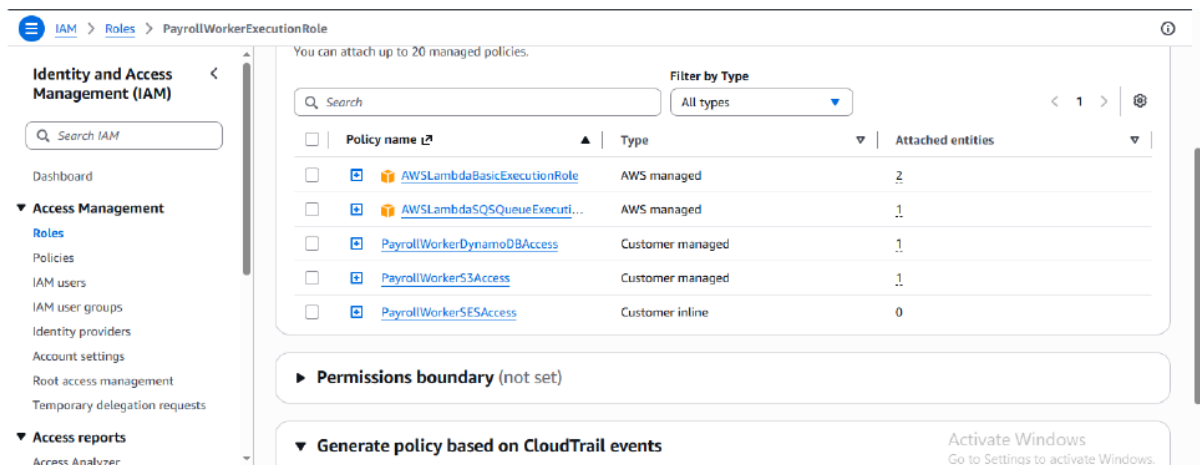


Appendix Figure D.2: S3 Payslip Object

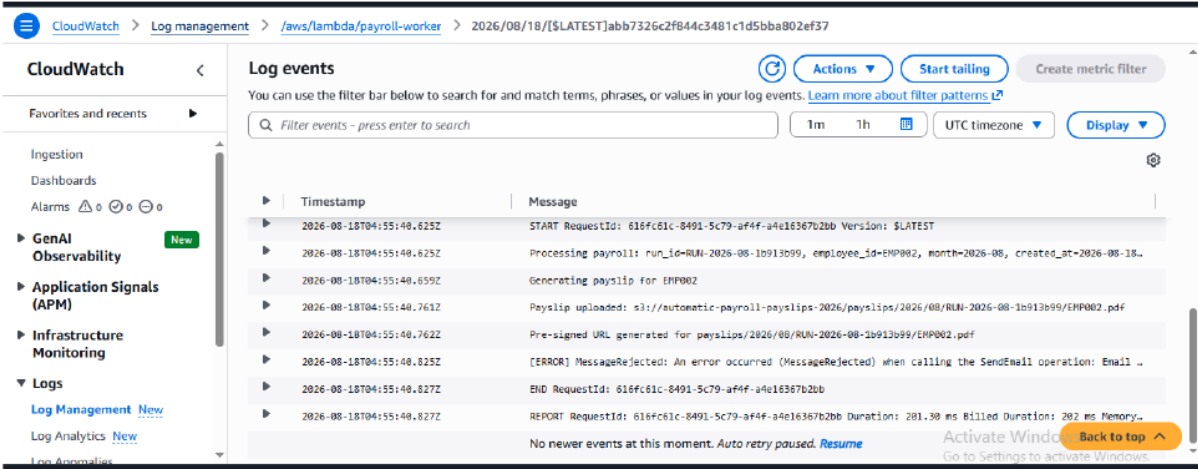


Appendix Figure D.3: Employee Payslip Email

Appendix E – Security and Monitoring Evidence



Appendix Figure E.1: IAM Execution Role



Appendix Figure E.2: CloudWatch Log